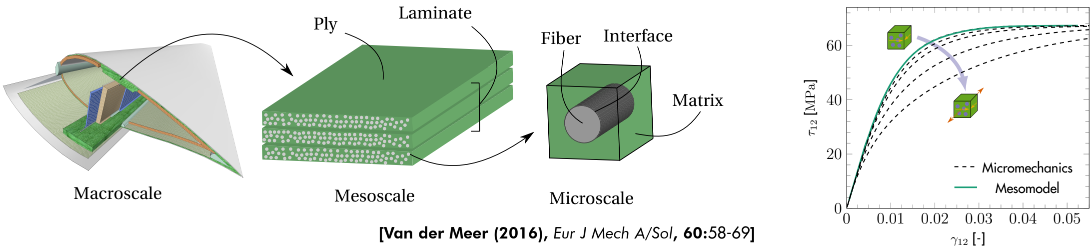
Modeling laminated composites

Numerical analysis of composites is an intrinsically multiscale endeavor

- Microscopic failure events $\Rightarrow$ macroscopic failure
- Upscaling: Mesomodel for a single ply $\Rightarrow$ lamination theory $\Rightarrow$ macromodel

Mesoscale models

- Efficiently upscale relevant nonlinear phenomena occurring at the microscale
- However, they can struggle to represent stress states not considered during calibration



[Van der Meer (2016), *Eur J Mech A/Sol*, 60:58-69]

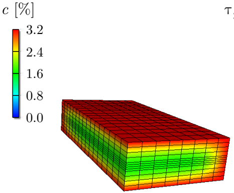
Concurrent multiscale modeling

Computational Homogenization (FE^2):

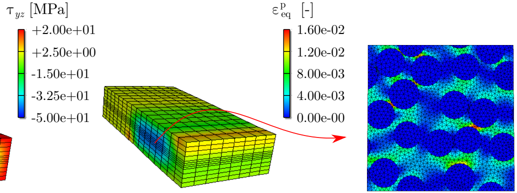
- A nested micromodel at every mesoscopic integration point
- General stress states and complex nonlinearities are captured without loss of generality

Drawback: Extreme computational effort

- A nonlinear BVP at every IP for every iteration of every time step of the mesomodel
- Execution time can easily reach the order of weeks even for simple problems



[Rocha et al. (2017), *Int J Num Meth Eng*, **112:**360-379**]**



[Rocha et al. (2019), *Eur J Mech A/Sol*, **73:**407-419**]**

Artificial neural networks

A regression model inspired by biological neural networks:

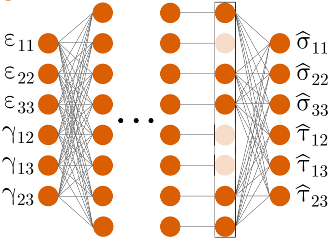
- A neural message is passed along a network of neurons through synaptic connections
- Neuron: Activated linear combination of values from previous layer

Dropout regularization:

- Neuron values are randomly dropped before reaching the final layer
- Provides robustness to small fluctuations and avoids overfitting

Use as micromodel surrogate

- Stresses: Forward propagation of macroscopic strains
- Stiffness: Backwards differentiation



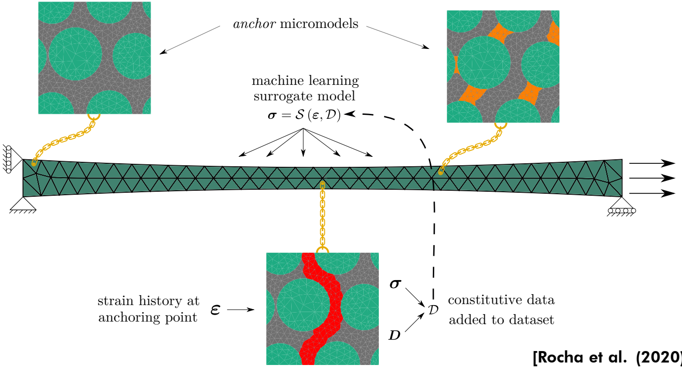
Drawback: no feedback on level of confidence

Offline
Physics
Data
Online

Active learning with Gaussian Processes

Learning constitutive behavior on the fly:

- Eliminates the need for sampling enormous parameter spaces offline
- Clustering for initial anchors, more added on demand at points with high uncertainty



[Rocha et al. (2020), *arXiv*:2007.07749**]**

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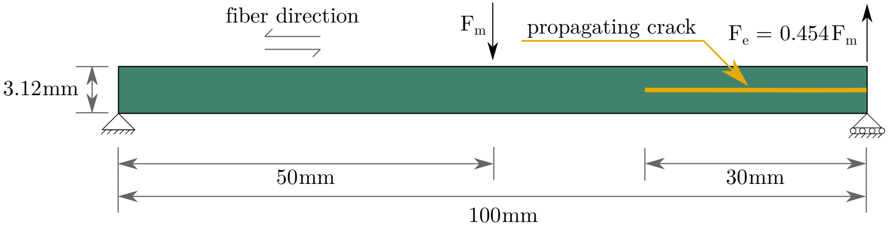
Crack propagation example

Using the same framework to approximate a traction-separation law:

- Mixed-mode bending specimen, orthotropic bulk and shifted cohesive zone law



[Rocha et al. (2020), *arXiv*:2007.07749**]**